USER INTERFACE DESIGN

EXPERIMENT 3: Difference between CLI(Command line interface) and GUI(Graphical user interface) and VUI (voice userinterface)

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Aim

To design and implement a file renaming system using Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI), and compare their usability and efficiency.

REQUIREMENTS

Hardware:

- Laptop/Desktop
- Microphone

Software:

- Python 3.x
- Tkinter Library
- SpeechRecognition
- PyAudio

Command Line Interface (CLI)

A Command Line Interface allows users to interact with the system by typing commands into a terminal.

Advantages:

- Fast execution
- Low memory usage
- Suitable for advanced users

Disadvantages:

- Requires knowledge of commands
- Not user-friendly for beginners

Implementation:

The CLI program renames files using command-line arguments.

Working:

- Accepts old filename and new filename as inputs
- Uses os.rename() function
- Displays success or error messages

```
import os

import sys
```

```python
def rename_file(old_name, new_name):

    try:

        os.rename(old_name, new_name)

        print(f"File renamed from {old_name} to {new_name}")

    except FileNotFoundError:

        print(f"Error: {old_name} not found.")

    except Exception as e:

        print(f"An error occurred: {e}")



if __name__ == "__main__":

    if len(sys.argv) != 3:

        print("Usage: python rename_file_cli.py <old_filename> <new_filename>")

    else:

        rename_file(sys.argv[1], sys.argv[2])
```

Output:

PS C:\Users\HP\Downloads\UID>python3 cli.py old.txt new.txt

File renamed from old.txt to new.txt

PS C:\Users\HP\Downloads\UID>

Graphical User Interface (GUI)

A GUI allows users to interact through graphical elements such as windows, buttons, and text fields.

Advantages:

- Easy to learn
- Interactive
- Reduces user errors

Disadvantages:

- Consumes more system resources
- Slower than CLI

Implementation:

The GUI application is built using Tkinter, allowing users to rename files through a graphical window.

Features:

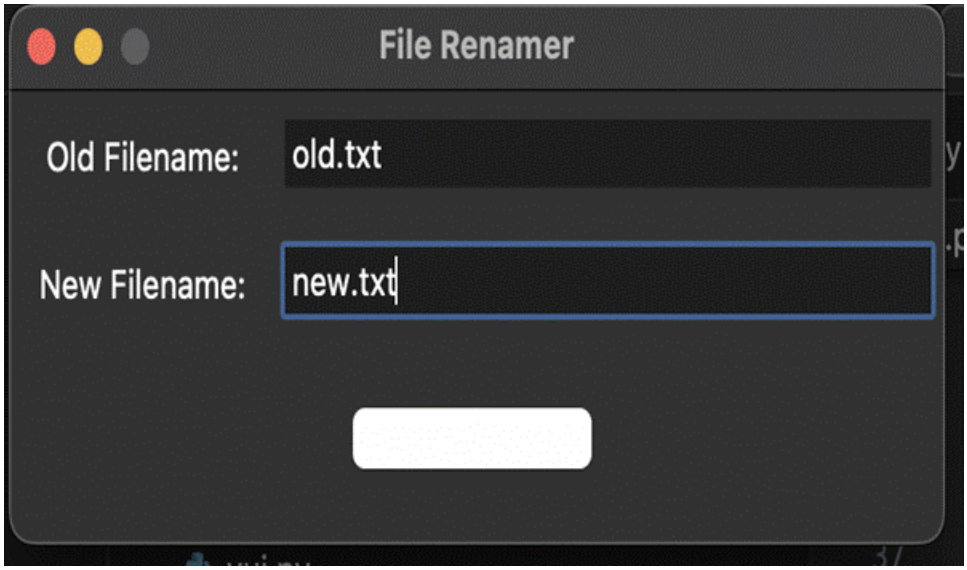
- Input fields for old and new filenames
- Rename button
- Popup messages for success or errors

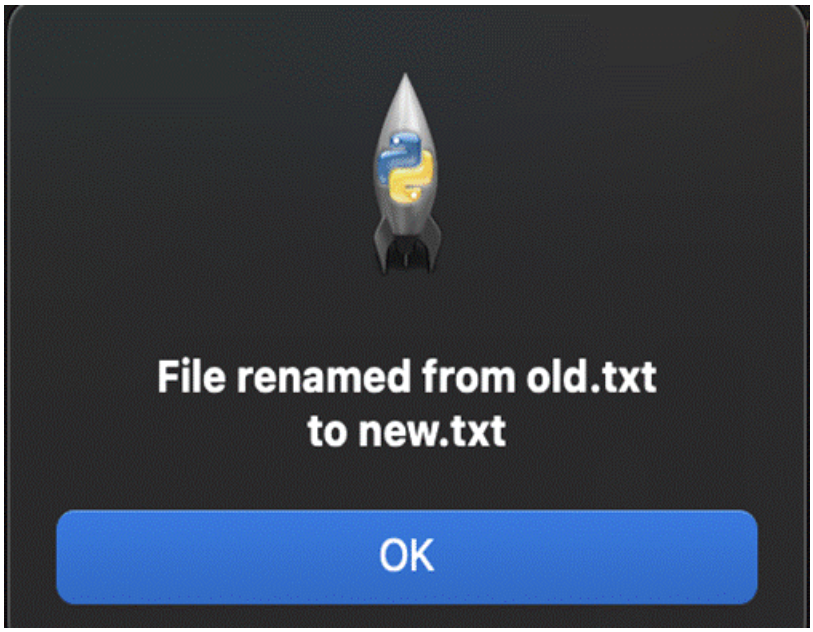
```
import tkinter as tk

from tkinter import messagebox
```

```python
import os


def rename_file():
    old_name = old_filename_entry.get()
    new_name = new_filename_entry.get()

    if not old_name or not new_name:
        messagebox.showwarning("Warning", "Please enter both filenames.")
        return

    try:
        os.rename(old_name, new_name)
        messagebox.showinfo("Success", f"File renamed from {old_name} to {new_name}")
    except FileNotFoundError:
        messagebox.showerror("Error", f"File '{old_name}' not found.")
    except Exception as e:
        messagebox.showerror("Error", f"An error occurred: {e}")


root = tk.Tk()
root.title("File Renamer")
root.geometry("400x150")   # makes window look professional
root.resizable(False, False)


tk.Label(root, text="Old Filename:").grid(row=0, column=0, padx=10, pady=10)
```

```
tk.Label(root, text="New Filename:").grid(row=1, column=0, padx=10, pady=10)



old_filename_entry = tk.Entry(root, width=30)

old_filename_entry.grid(row=0, column=1)



new_filename_entry = tk.Entry(root, width=30)

new_filename_entry.grid(row=1, column=1)



rename_button = tk.Button(root, text="Rename File", bg="green", fg="white",
command=rename_file)

rename_button.grid(row=2, columnspan=2, pady=15)



root.mainloop()
```

Output:





Voice User Interface (VUI)

A Voice User Interface enables interaction using voice commands and speech recognition technology.

Advantages:

- Hands-free operation
- Improves accessibility
- Natural communication

Disadvantages:

- Dependent on voice clarity
- Sensitive to background noise

Implementation:

The VUI system uses speech recognition to rename files based on voice commands.The spoken command is converted into text, and the file is renamed using Python.

```
import speech_recognition as sr

import os



recognizer = sr.Recognizer()



try:

   with sr.Microphone() as source:

       print("Say: rename old.txt to new.txt")
```

```
        recognizer.adjust_for_ambient_noise(source)

        audio = recognizer.listen(source)


    try:

        command = recognizer.recognize_google(audio)

        print("You said:", command)


        words = command.lower().split()


        # Common speech recognition corrections

        corrections = {

                "text": "txt",

                "pdf": "pdf",

                "doc": "doc"

        }


        # Reconstruct filenames that were split by "dot"

        # Find indices where "dot" appears and merge them

        reconstructed = []

        i = 0

        while i < len(words):

                if i + 2 < len(words) and words[i + 1] == "dot":

                # Merge word + dot + extension

                extension = words[i + 2]
```

```
            # Apply corrections to extension
            if extension in corrections:
                extension = corrections[extension]
            reconstructed.append(words[i] + "." + extension)
            i += 3
            else:
            reconstructed.append(words[i])
            i += 1

    # Now extract old and new filenames
    # Should be: rename <old_file> to <new_file>
    old_name = reconstructed[1]
    new_name = reconstructed[3]

    os.rename(old_name, new_name)
    print("File renamed successfully!")

except sr.UnknownValueError:
    print("Error: Could not understand audio")
except sr.RequestError as e:
    print(f"Error: API request failed - {e}")
except Exception as e:
    print("Error:", e)
```

```
except OSError as e:

    print(f"Error: Microphone not available - {e}")

    print("Please ensure a microphone is connected and properly configured.")
```

Output:

PS C:\Users\HP\Downloads\UID>python3 vui.py

Say: rename old.txt to new.txt

You said: rename old dot text to new dot text

File renamed successfully!

PS C:\Users\HP\Downloads\UID>

RESULT:

The file renaming system was successfully implemented using CLI, GUI, and VUI. Each interface allowed users to rename files effectively while demonstrating different levels of usability.